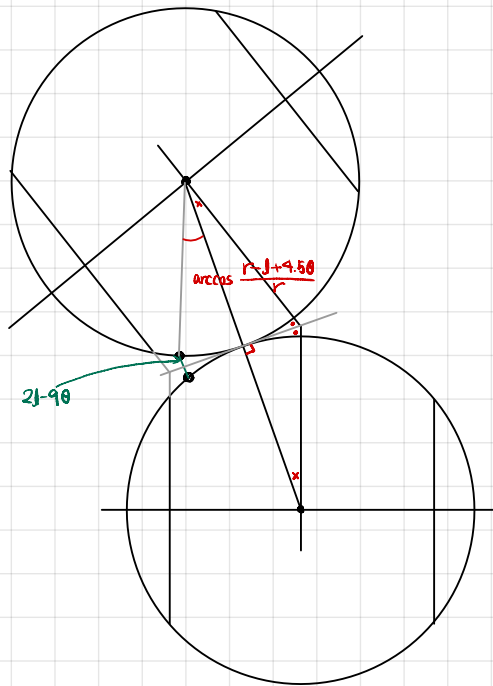
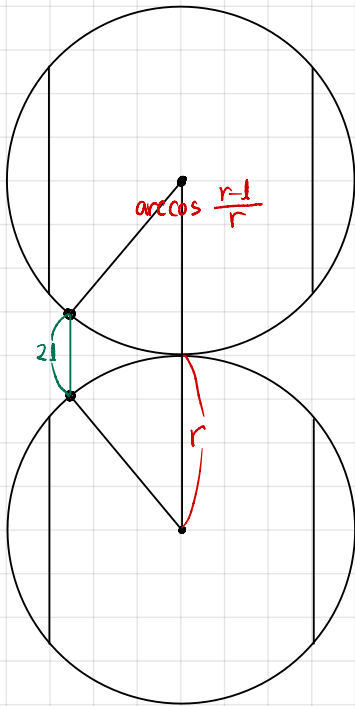


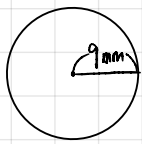
0만큼 회전
→ 실은 90mm 당김.

영지 pip, dip



$$= \frac{\pi}{2} - \arccos \frac{r-1}{r} + \arccos \frac{r-1+4.50}{r}$$

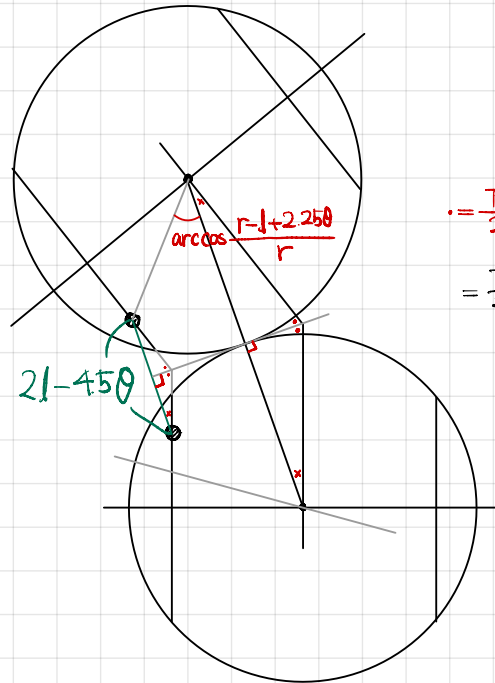
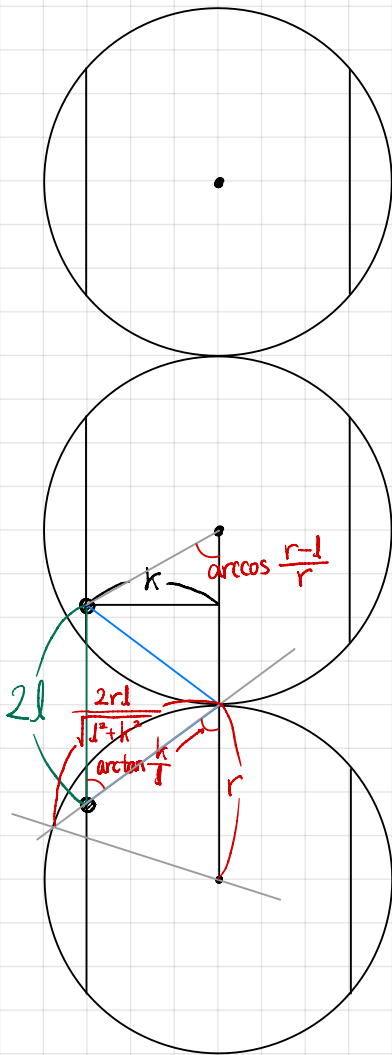
$$= \frac{\pi}{2} - \arccos \frac{3.6}{7.6} + \arccos \frac{3.6+4.50}{7.6}$$



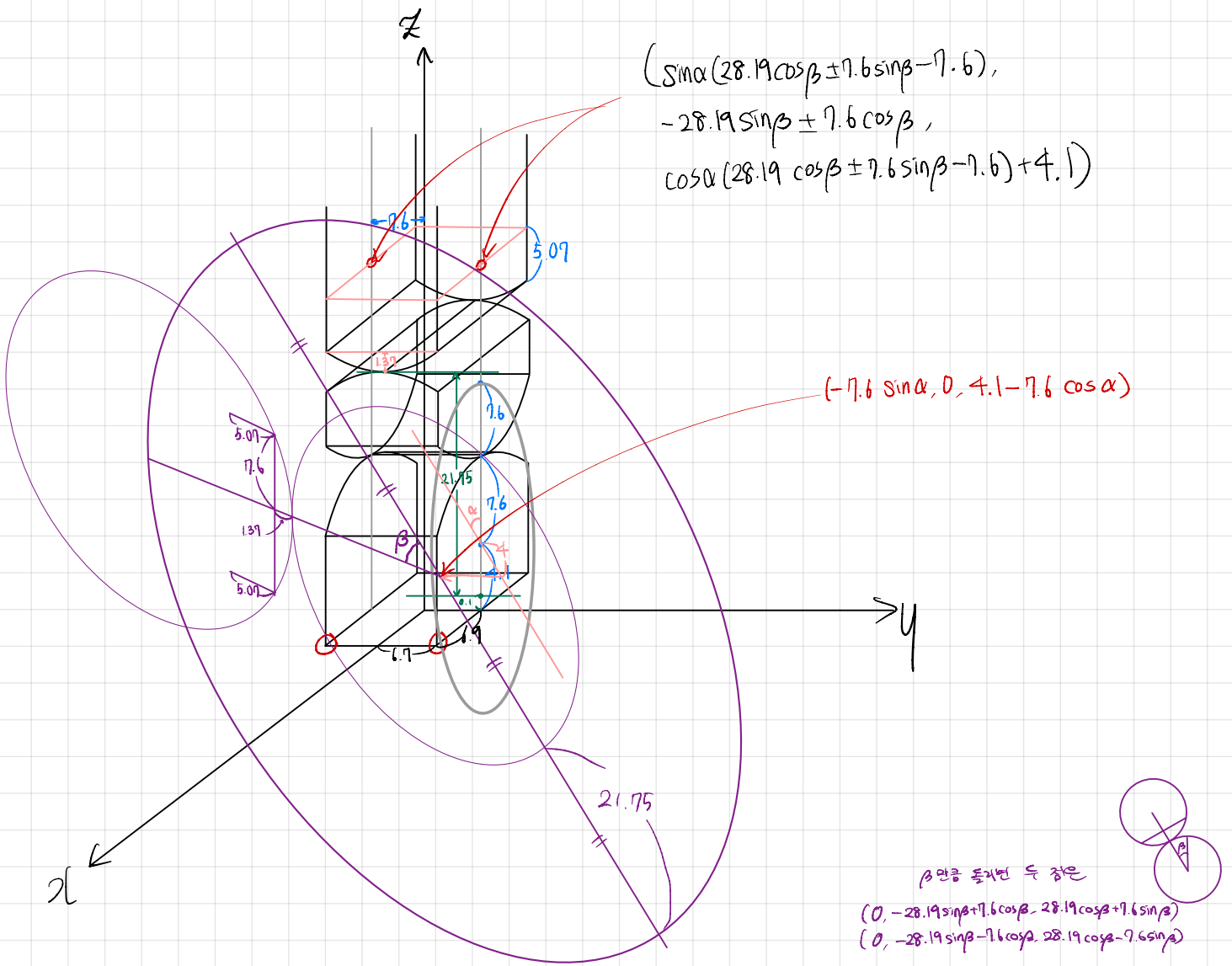
9만큼 회전
→ 실은 90mm 당김.

dip

$$c^2 = a^2 + b^2 - 2ab \cos C$$



$$\begin{aligned} &= \frac{\pi}{2} - \arccos \frac{r-1}{r} + \arccos \frac{r-1+2.250}{r} \\ &= \frac{\pi}{2} - \arccos \frac{2.6}{7.6} + \arccos \frac{2.6+2.250}{7.6} \end{aligned}$$



$$\begin{aligned} & \{ \sin \alpha (28.19 \cos \beta + 7.6 \sin \beta - 7.6) - 6.9 \}^2 + \{ -28.19 \sin \beta + 7.6 \cos \beta - 6.9 \}^2 \\ & + \{ \cos \alpha (28.19 \cos \beta + 7.6 \sin \beta - 7.6) + 4.1 \}^2 \end{aligned}$$

$$\begin{aligned} &= 7.6^2 [(A + \sin \alpha \sin \beta)^2 + (B + \cos \beta)^2 + (C + \cos \alpha \sin \beta)^2] \\ &= 7.6^2 [A^2 + 2A \sin \alpha \sin \beta + B^2 + 2B \cos \beta + C^2 + 2C \cos \alpha \sin \beta + 1] = 9^2 (32 - 0)^2 \end{aligned}$$

$$\begin{aligned} & \{ \sin \alpha (28.19 \cos \beta - 7.6 \sin \beta - 7.6) - 6.9 \}^2 + \{ -28.19 \sin \beta - 7.6 \cos \beta - 6.9 \}^2 \\ & + \{ \cos \alpha (28.19 \cos \beta - 7.6 \sin \beta - 7.6) + 4.1 \}^2 \end{aligned}$$

$$\begin{aligned} &= 7.6^2 [(A - \sin \alpha \sin \beta)^2 + (B - \cos \beta)^2 + (C - \cos \alpha \sin \beta)^2] \\ &= 7.6^2 [A^2 - 2A \sin \alpha \sin \beta + B^2 - 2B \cos \beta + C^2 - 2C \cos \alpha \sin \beta + 1] = 9^2 (3.2 + 0)^2 \end{aligned}$$

$$\frac{1}{2} \times \left(\frac{9}{7.6} \right)^2 \{ (\theta_1 - 3.2)^2 + (\theta_2 + 3.2)^2 \} = A^2 + B^2 + C^2 + 1$$

$$\begin{aligned} \frac{1}{4} \times \left(\frac{9}{7.6} \right)^2 (\theta_1 - \theta_2 - 6.4)(\theta_1 + \theta_2) &= A \sin \alpha \sin \beta + B \cos \beta + C \cos \alpha \sin \beta = \sin \beta (A \sin \alpha + C \cos \alpha) + B \cos \beta \\ &= \sin \beta (0.5 \cos \alpha - 0.9 \sin \alpha + 3.7 \cos \beta - 1) - 3.7 \cos \beta \sin \beta - 0.9 \cos \beta \end{aligned}$$

$$\begin{aligned} A &= 3.7 \sin \alpha \cos \beta - \sin \alpha - 0.9 \\ B &= -3.7 \sin \beta - 0.9 \\ C &= 3.7 \cos \alpha \cos \beta - \cos \alpha + 0.5 \end{aligned}$$